LAB EXPERIMENT RECORD

SET 13 – GEOGRAPHIC DATA VISUALIZATION

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Date	8 October 2026
Day	Thursday

Aim

To analyze geographic data using a map chart, scatter plot, table, and an interactive dashboard in RStudio.

Dataset

City	Population	Avg. Temperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

Note

The dataset contains City A, City B, and City C without real geographic names or coordinates. For the map visualization, illustrative coordinates are used only to demonstrate the mapping technique.

PROGRAM 1: Map Chart – Geographic Distribution

To create a map chart to visualize the distribution of cities on a geographic map and label the map elements.

Required Packages

```
install.packages("ggplot2")
install.packages("maps")
```

R Program

```
library(ggplot2)
library(maps)

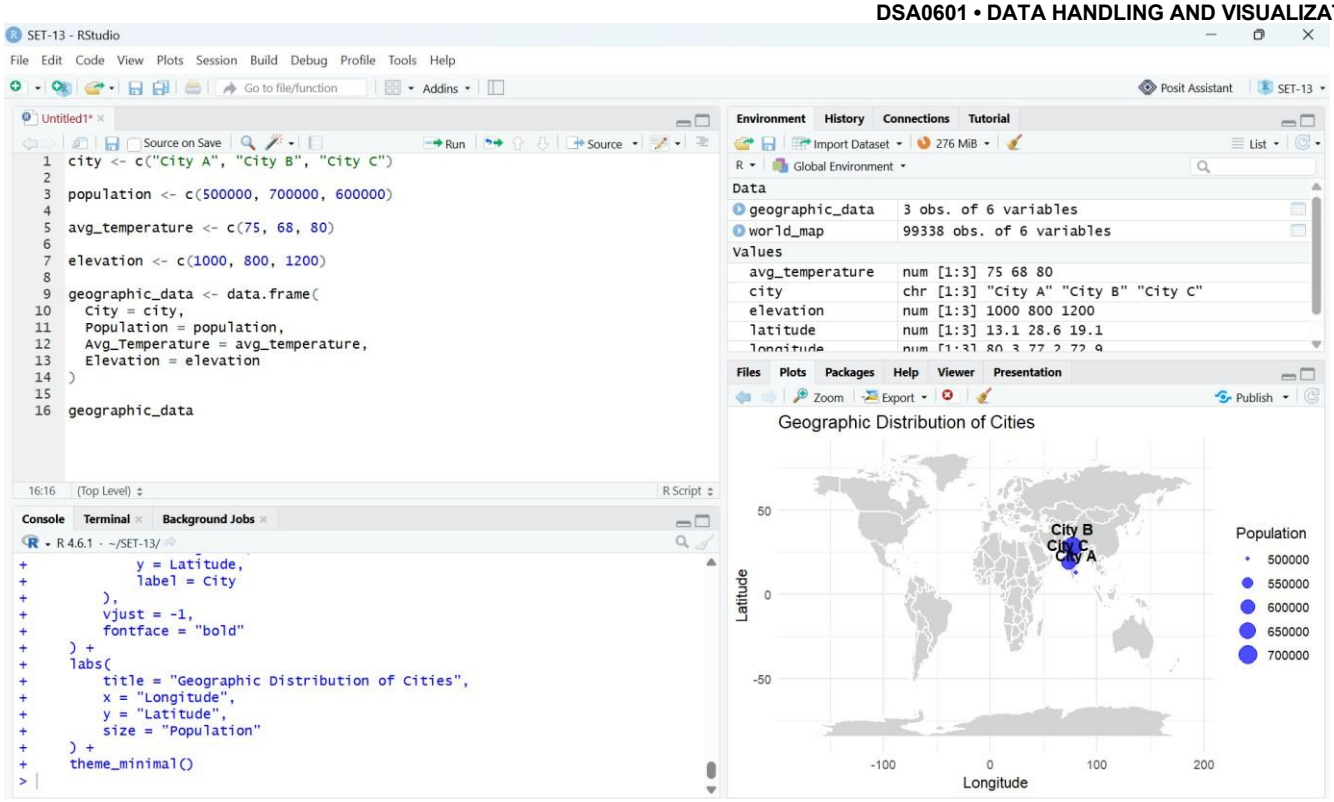
city <- c("City A", "City B", "City C")
population <- c(500000, 700000, 600000)
avg_temperature <- c(75, 68, 80)
elevation <- c(1000, 800, 1200)

# Illustrative coordinates because the dataset gives placeholder city names
longitude <- c(80.27, 77.21, 72.88)
latitude <- c(13.08, 28.61, 19.07)

geographic_data <- data.frame(
  City = city,
  Population = population,
  Avg_Temperature = avg_temperature,
  Elevation = elevation,
  Longitude = longitude,
  Latitude = latitude
)

world_map <- map_data("world")

ggplot() +
  geom_polygon(
    data = world_map,
    aes(x = long, y = lat, group = group),
    fill = "lightgray",
    color = "white"
  ) +
  geom_point(
    data = geographic_data,
    aes(x = Longitude, y = Latitude, size = Population),
    color = "blue",
    alpha = 0.7
  ) +
  geom_text(
    data = geographic_data,
    aes(x = Longitude, y = Latitude, label = City),
    vjust = -1,
    fontface = "bold"
  ) +
  labs(
    title = "Geographic Distribution of Cities",
    x = "Longitude",
    y = "Latitude",
    size = "Population"
  ) +
  theme_minimal()
```



Output 1: Geographic Distribution of Cities – RStudio Execution Screenshot

Observation: The map displays City A, City B, and City C as labelled points. Point size represents population. City B has the largest population (700,000), followed by City C (600,000) and City A (500,000).

PROGRAM 2: Scatter Plot – Average Temperature vs Population

To generate a scatter plot to explore the relationship between average temperature and population size.

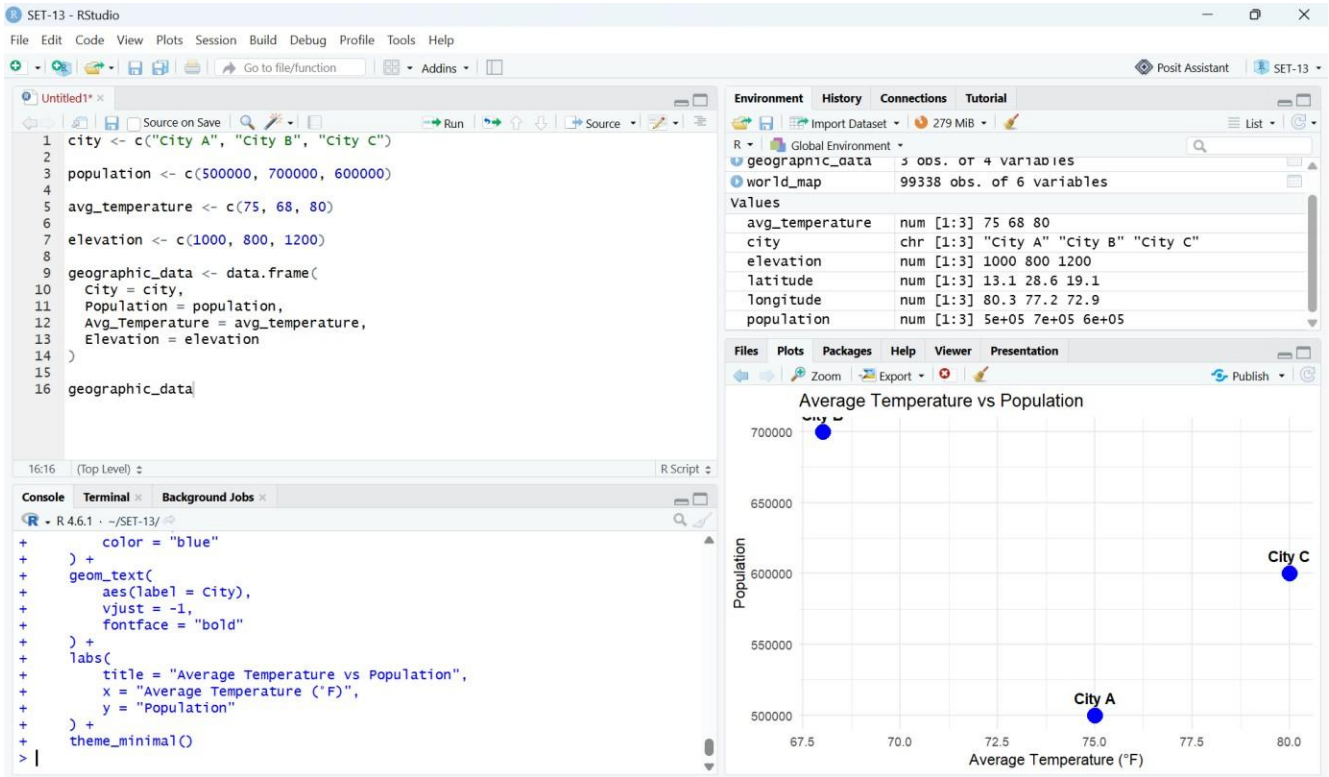
R Program

```
library(ggplot2)

city <- c("City A", "City B", "City C")
population <- c(500000, 700000, 600000)
avg_temperature <- c(75, 68, 80)
elevation <- c(1000, 800, 1200)

geographic_data <- data.frame(
  City = city,
  Population = population,
  Avg_Temperature = avg_temperature,
  Elevation = elevation
)

ggplot(
  geographic_data,
  aes(x = Avg_Temperature, y = Population)
) +
  geom_point(size = 5, color = "blue") +
  geom_text(
    aes(label = City),
    vjust = -1,
    fontface = "bold"
  ) +
  labs(
    title = "Average Temperature vs Population",
    x = "Average Temperature (°F)",
    y = "Population"
  ) +
  theme_minimal()
```



Output 2: Average Temperature vs Population – RStudio Execution Screenshot

Observation: City B has the highest population (700,000) but the lowest average temperature (68°F). City C has the highest temperature (80°F) with a population of 600,000, while City A has 75°F and 500,000. With only three observations, the data does not provide strong evidence of a consistent relationship between temperature and population.

PROGRAM 3: Geographic Data Table

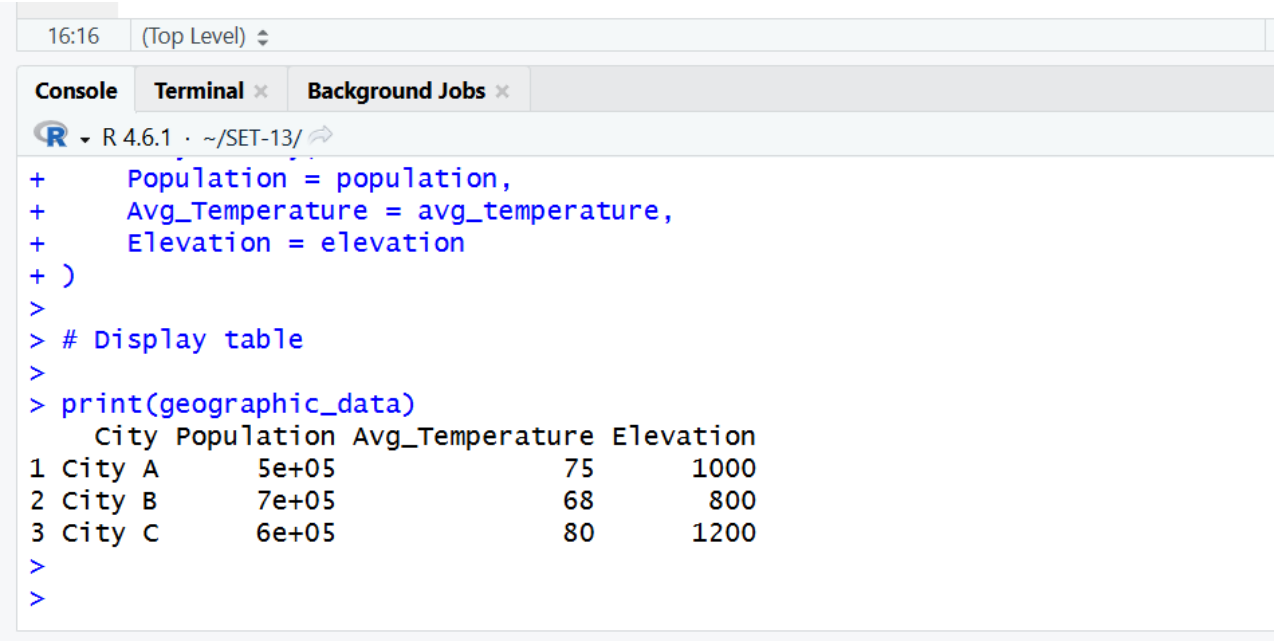
To build a table to display the geographic data for each city and label the table elements.

R Program

```
city <- c("City A", "City B", "City C")
population <- c(500000, 700000, 600000)
avg_temperature <- c(75, 68, 80)
elevation <- c(1000, 800, 1200)

geographic_data <- data.frame(
  City = city,
  Population = population,
  Avg_Temperature = avg_temperature,
  Elevation = elevation
)

print(geographic_data)
```



Output 3: Geographic Data Table – RStudio Console Screenshot

Observation: The table displays City, Population, Average Temperature, and Elevation for all three records. It provides exact numerical values for detailed comparison.

PROGRAM 4: Interactive Dashboard in RStudio

To develop an interactive dashboard combining the map chart, scatter plot, and geographic data table. The dashboard is implemented using Shiny in RStudio.

R Program

```
library(shiny)
library(ggplot2)
library(maps)

city <- c("City A", "City B", "City C")
population <- c(500000, 700000, 600000)
avg_temperature <- c(75, 68, 80)
elevation <- c(1000, 800, 1200)

longitude <- c(80.27, 77.21, 72.88)
latitude <- c(13.08, 28.61, 19.07)

geographic_data <- data.frame(
  City = city,
  Population = population,
  Avg_Temperature = avg_temperature,
  Elevation = elevation,
  Longitude = longitude,
  Latitude = latitude
)

world_map <- map_data("world")

ui <- fluidPage(
  titlePanel("Geographic Data Dashboard"),

  sidebarLayout(
    sidebarPanel(
      selectInput(
        "selected_city",
        "Select City:",
        choices = c("All Cities", city),
        selected = "All Cities"
      )
    ),

    mainPanel(
      fluidRow(
        column(
          6,
          h3("City Distribution Map"),
          plotOutput("map")
        ),
        column(
          6,
          h3("Temperature vs Population"),
          plotOutput("scatter")
        )
      ),
      hr(),
      h3("Geographic Data Table"),
      tableOutput("table")
    )
  )
)

server <- function(input, output) {

  filtered_data <- reactive({
    if (input$selected_city == "All Cities") {
```

```
    geographic_data
  } else {
    geographic_data[
      geographic_data$City == input$selected_city,
    ]
  }
})

output$map <- renderPlot({
  data <- filtered_data()

  ggplot() +
    geom_polygon(
      data = world_map,
      aes(x = long, y = lat, group = group),
      fill = "lightgray",
      color = "white"
    ) +
    geom_point(
      data = data,
      aes(x = Longitude, y = Latitude, size = Population),
      color = "blue",
      alpha = 0.7
    ) +
    geom_text(
      data = data,
      aes(x = Longitude, y = Latitude, label = City),
      vjust = -1,
      fontface = "bold"
    ) +
    labs(
      title = "Geographic Distribution of Cities",
      x = "Longitude",
      y = "Latitude",
      size = "Population"
    ) +
    theme_minimal()
})

output$scatter <- renderPlot({
  data <- filtered_data()

  ggplot(
    data,
    aes(x = Avg_Temperature, y = Population)
  ) +
    geom_point(size = 5, color = "blue") +
    geom_text(
      aes(label = City),
      vjust = -1,
      fontface = "bold"
    ) +
    labs(
      title = "Average Temperature vs Population",
      x = "Average Temperature (°F)",
      y = "Population"
    ) +
    theme_minimal()
})

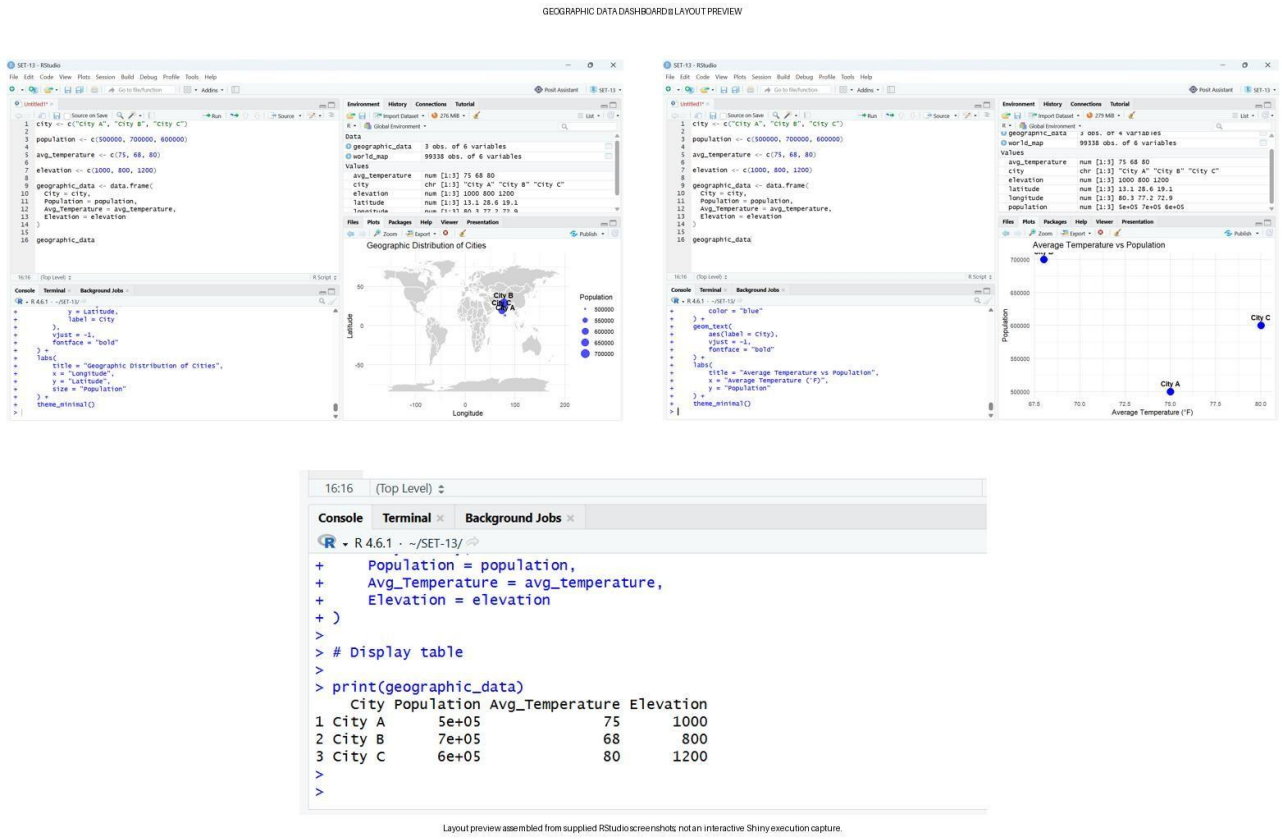
output$table <- renderTable({
  data <- filtered_data()

  data.frame(
    City = data$City,
    Population = data$Population,
    Avg_Temperature = data$Avg_Temperature,
    Elevation = data$Elevation
  )
```

```
  })
}

shinyApp(ui = ui, server = server)
```

PROGRAM 4: Dashboard Output and Explanation



Output 4: Geographic Data Dashboard – Layout Preview

Observation: The dashboard combines the geographic map, temperature-versus-population scatter plot, and data table. The city selector filters the displayed information so that the user can explore All Cities or an individual city.

The preview above is assembled from the supplied RStudio screenshots. It is not a screenshot of an executed Shiny application. After running the dashboard with Run App, capture the actual Shiny application and replace this preview if execution evidence is required.

Procedure

• Create vectors for City, Population, Average Temperature, and Elevation.

• Combine the vectors using data.frame() to create the geographic dataset.

• Install and load ggplot2 and maps for geographic visualization.

• Assign illustrative latitude and longitude values because the supplied city names do not have actual coordinates.

• Create the geographic map using geom_polygon(), geom_point(), and geom_text().

• Create the scatter plot using Avg_Temperature on the x-axis and Population on the y-axis.

• Create the geographic data table using data.frame() and print().

• Install and load Shiny for the interactive dashboard.

• Create the dashboard interface using fluidPage(), sidebarLayout(), selectInput(), plotOutput(), and tableOutput().

• Use reactive() to filter the map, scatter plot, and table when a city is selected.

• Run the Shiny application and verify all three views.

Overall Insights

• City B has the largest population at 700,000.

• City C has the highest average temperature at 80°F.

• City C also has the highest elevation at 1,200 m.

• City B has the lowest average temperature at 68°F and the lowest elevation at 800 m.

• The scatter plot does not establish a reliable temperature-population relationship because the dataset contains only three observations.

• The map provides spatial-style visual comparison, while the scatter plot shows the relationship between two numerical variables.

• The table provides exact values and supports precise comparison.

• The Shiny dashboard combines all three views and allows the user to select a city interactively.

Result

The geographic data was successfully analyzed in RStudio using a map chart, scatter plot, and data table. An interactive Shiny dashboard was developed to combine the three views for geographic data exploration.

Important Submission Note

The supplied RStudio screenshots were inserted into the corresponding sections. The map screenshot uses illustrative coordinates because the original dataset does not provide real city locations. For final dashboard evidence, use a screenshot of the Shiny application after clicking Run App if your faculty requires proof of interactive execution.